

Module 7 — Observability (Orbit Academy v5.5)

Printable edition. The interactive version — live simulator, self-checking questions — is the course website; open the matching tut/worksheet pages there. Dynamic diagrams appear here as bracketed figure notes.

What you will learn

- how **active radar** finds objects — and why its echo weakens as the **fourth power of range**, so LEO is easy, GEO is hard, and xGEO is nearly out of reach.
- why optical tracking only works on **sunlit** objects (the “driving at night by a truck’s headlights” analogy), how **thermal-IR** lets warm objects be seen even in shadow, and the difference between **cooperative** and **uncooperative** objects.
- what a radar measures (**range** and **range-rate**, crude angles) versus what an optical sensor measures (**precise angles — RA/DEC — but no range**), and how **parallax** recovers the missing range.
- why keeping **custody** of an object demands re-observation at a **cadence** set by how fast its orbit goes stale — and how that enables **maneuver detection**.
- the ladder from **orbitology** (where is it?) to **characterization** (what is it?) to **inferring intent** (what is it doing?), and the split between wide **surveys** and pointed **custody** observations.

Quick-start tutorial

Every earlier module assumed we already **knew** where a satellite was. This one asks the harder question that underlies all of space domain awareness: **how do we find and track objects in the first place, and what can we actually learn about them?**

Start in the optical: what can we measure in an image?

The simplest sensor is a telescope taking a picture. Look at that image of a satellite and ask: **what, precisely, can we measure?** Just two things — its **position** and its **brightness**. Everything optical flows from those two numbers.

Position is measured *relative to the background stars*, whose directions are known to exquisite precision. We report it as two angles — **Right Ascension (RA)** and **Declination (Dec)** — which are just **Earth’s longitude and latitude projected up onto the sky** (the equator and poles extended outward become the celestial equator and poles). A single image therefore gives a precise **direction** to the object, but (by itself) **no distance**.

Brightness is measured in **astronomical magnitudes** (also called **apparent magnitudes**). Magnitudes are a backwards, logarithmic scale worth memorizing semi-quantitatively: **bigger numbers mean fainter** objects — and the very brightest objects have **negative** magnitudes. The one fact to remember: **5 magnitudes is a factor of 100** in brightness (so each step of 1 magnitude is a factor of $\sqrt[5]{100} \approx 2.512$, a ratio formalized by Norman Pogson in 1856), and **10 magnitudes is $100 \times 100 = 10,000\times$** . The zero point was traditionally anchored to the star **Vega** ($m \approx 0$) in the constellation Lyra; since Vega turns out to be slightly variable, the modern scale is tied to calibrated

physical flux standards instead — but “about as bright as Vega” remains the right mental picture for $m = 0$.

Notation: the official symbol for apparent magnitude is a lowercase **m**. A capital **M** means something different — **absolute magnitude**, a distance-independent measure of an object’s intrinsic luminosity that we do not use in this course. Beware: the loose abbreviation “**mag**” gets used for *both* quantities, so when a colleague quotes “mag 7” it pays to ask which one they mean. In this course, magnitudes are always apparent magnitudes, written **m**.

m	object (brightest → faintest)
–26.7	the Sun — never point a telescope at it
–4.4	Venus near its brightest
–1.46	Sirius — the brightest star in the night sky
0	Vega (constellation Lyra) — the traditional zero point
+1.98	Polaris , the North Star — a Cepheid variable whose light pulses by a few hundredths of a magnitude every ~4 days
+4	this module’s satellite at its sharp one-sided glint
+6	naked-eye limit under a dark sky (anything fainter needs optics)
+8.5	the sim’s GEO satellite, ordinary fully-lit face
+14	the same satellite on its back-lit near-dark pass
+30	roughly the faintest objects the Hubble Space Telescope can detect

How bright a sunlit satellite *looks* depends on many things at once: the **illumination phase** (Sun-satellite-observer angle), the **orientation** of the **bus** — the satellite’s central body, the box that houses the payload and electronics, off which the solar panels and antennas hang — and of the solar panels themselves (a flat panel can throw a brief mirror-like **glint**), the **distance**, and the **albedo** — the fraction of light a surface reflects (shiny foil ≈ 1 , dull black ≈ 0). Change any one and the brightness changes. The simulator’s **light curve** shows this: brightness spans ~10 magnitudes (10,000 $\times$) from the sharp one-sided **glint** ($m \approx 4$) to the back-lit near-dark pass ($m \approx 14$) — the ordinary fully-lit face ($m \approx 8.5$) sits between those extremes.

There are only a few ways to sense a distant object. Besides collecting **reflected sun-light** (optical, above), we can **bounce our own radio waves off it** (active radar) or **collect the heat it emits** (thermal infrared). Each has hard physical limits — and understanding those limits is what separates “we see everything” from the truth: much of space is genuinely hard to watch.

Radar: shout and listen for the echo

A radar sends out a pulse and listens for the faint echo. Because the pulse spreads out on the way **there** and the echo spreads out on the way **back**, the received power falls as the **fourth power of range** — double the distance and the echo is **16 $\times$** weaker. This one fact explains why radar tracks LEO easily, struggles at GEO, and mostly gives up in cislunar space.

Picture this: Optical tracking is like **driving a dark road with your head-lights off**. You cannot see bicycles (satellites) on their own — they emit no

light. But when a **truck (the Sun)** comes along, its headlights catch the bicycles' reflectors and suddenly you can see them. You are not lighting them; you are catching **reflected** light. So you can only see a satellite when the Sun is lighting it *and* your sky is dark enough to see it — the same geometry that gives the Moon its phases.

A satellite in **Earth's shadow** reflects no sunlight and vanishes to an optical sensor — but it is still warm, so a **thermal-infrared** sensor can catch its own glow. Radar is independent of sunlight entirely: it is an **active** sensor that transmits its own radio-frequency pulses and detects the reflected echo — but it pays the range⁴ penalty.

What each sensor actually measures

This is the crux. **Radar** times the echo's round trip, so it measures **range** (distance) precisely and, from the Doppler shift, **range-rate** (how fast the distance is changing) — but only **crude angles**. **Optical** is the opposite: a telescope pins the object's **direction** on the sky very precisely (its **angles**, written as Right Ascension and Declination, RA/DEC), but a single image gives **no distance at all**. To recover range from optical, you use **parallax** — the shift in apparent direction seen from two vantage points (or the same telescope at two times as the Earth carries it along).

Picture this: Parallax demo you can do right now: hold a finger at arm's length and look at it with one eye, then the other. Your finger jumps against the background. Bring it closer — it jumps **more**. That jump is parallax, and its size tells you the distance. Two telescopes (or one at two times) do exactly this to a satellite.

Custody: it is not enough to find it once

Finding an object once is easy; **keeping** track of it — maintaining **custody** — is the hard part, because predictions go stale. In simple LEO an orbit stays predictable for a long time; in the chaotic cislunar regime (Module 6) a small "where is it now" error explodes fast, so you must re-observe at a brisk **cadence**. Comparing a new observation against the predicted position is also how you **detect a maneuver**: if the object is not where its old orbit says it should be, it performed a **propulsive maneuver — a burn**. Note what the observation does and does not tell you: you know its **velocity changed**, but not *why* — it might be routine station-keeping, a relocation, a move toward disposal, or something less benign. Sorting that out takes continued watching (and is exactly the "intent" problem below).

Finally, the questions you can ask about an object form a **ladder**, each rung more ambitious — and more data-hungry — than the one below. **Orbitology** answers *where is it and where is it going*. **Characterization** answers *what is it* — size, shape, spin, whether it has a payload or a grappling arm. **Inferring intent** is the hardest and most consequential: *what is it doing, and why* — is that close approach an inspection, a docking, or a threat? Wide **all-sky surveys** keep a catalog of everything that moves; pointed **custody observations** stare hard at the few objects that matter. This module is about all of it.

PART A · Illumination: seeing by reflected sunlight

Why you can only see a satellite the way you see a bicycle at night — when something else lights it. (Sim scene: GEO phases.)

Exercise 1 — The headlight analogy — you see reflected sunlight

An optical telescope is **passive**: it collects light, it does not send any. A satellite makes no visible light of its own, so the only way to see it optically is by the **sunlight it reflects** — exactly how you see the Moon, or a bicycle reflector.

Picture driving a dark road with your headlights off. You cannot see the bicycles around you. Then a **truck** comes by — its headlights catch the bikes' reflectors and they light up. The **truck is the Sun**; the **bicycles are satellites**; you are the observer catching reflected light. You need the Sun to light the target *and* a dark sky to see it against.

Predict first: It's the middle of a bright afternoon. Can a ground optical telescope easily track a faint satellite overhead? Why or why not?

Think it through — no sim needed:

Think about the two conditions optical needs: the target lit by the Sun, and a dark background sky. Ask when both hold. (One caution about the GEO phases scene: its light-curve graph plots brightness against the Sun-satellite-observer **phase angle**, not against time of night — so twilight does not appear on that graph. Twilight lives in the *geometry*: it is the special time when the satellite, high up, is still in sunlight while your sky below has already gone dark.)

What to look for: both conditions are needed. The best time is twilight: the satellite up high is still in sunlight while the ground observer's sky has gone dark. In full daylight the bright sky drowns the faint reflection; in the dead of night a low satellite is in Earth's shadow.

While you watch, think about:

- Why is twilight (just after sunset / before sunrise) the prime time for optical satellite tracking?
- In the analogy, what real thing is the truck, and what is the bicycle reflector?
- Why can't a telescope simply "shine a light" on the satellite the way radar does?

Check your understanding: Why can an optical telescope only see a satellite under certain conditions?

- Satellites are simply too small ever to be seen from the ground
- It sees only reflected sunlight, so the target must be lit and the sky dark
- Optical telescopes can only be operated during the middle of the day
- Satellites emit their own visible light, which is swamped by daylight

Exercise 2 — Into the shadow — the satellite vanishes

Follow a satellite as it orbits into **Earth's shadow**. The moment the Earth blocks the Sun, the satellite stops reflecting sunlight and **disappears** to an optical sensor — even though nothing about the object itself changed. This is the same reason a whole cloud of Starlink satellites winks out as it crosses into night.

This shadow blind-spot is a real operational gap for optical-only tracking. It is one reason you want **multiple sensing methods**: radar and thermal-IR both keep working in the shadow.

Predict first: A satellite you are optically tracking is about to cross into Earth's shadow. Predict what your telescope sees at the moment it crosses.

Think it through — no sim needed:

Imagine watching a satellite approach the shadow boundary (the terminator) and cross it.

What to look for: the satellite fades and vanishes as it enters shadow — no sunlight to reflect. It is still there and still warm; only its reflected-light signature is gone. Radar or thermal-IR would still detect it.

While you watch, think about:

- Why does an object in Earth's shadow disappear optically but not to radar?
- How could an operator exploit shadow timing to reduce how often their satellite is optically observed?
- What sensor would you add to cover the shadow gap, and why?

Check your understanding: A satellite crosses into Earth's shadow. What happens to an optical tracker?

- It sees the satellite more clearly, with less glare to contend with
- It loses the target — no sunlight to reflect — though radar and IR still work
- Nothing changes, since the telescope detects the satellite directly
- The satellite is destroyed by the sudden thermal shock of the shadow

Exercise 3 — Thermal infrared — warm objects glow in the dark

Everything warmer than its surroundings **emits infrared light** of its own — this is thermal radiation, the same reason night-vision and **FLIR** (forward-looking infrared) cameras work. A satellite sits at tens of degrees; it glows faintly in the **thermal infrared** whether or not the Sun is on it.

So a thermal-IR sensor can catch an object **in Earth's shadow**, when reflected-sunlight optical goes blind. The catch: IR sensors usually must be **cooled** (so the sensor's own heat doesn't swamp the signal), and everything else glows too — Earth's atmosphere radiates strongly in the infrared (chiefly its **water vapor**, **carbon dioxide** and **ozone**), and so do the telescope's own warm mirrors and structure — so a faint satellite must be picked out against a bright, fluctuating IR background. Thermal IR is a powerful complement to reflected-light optical, but not a free lunch.

Predict first: A satellite is in Earth's shadow, invisible to a normal telescope. Is there any passive way (no radar) to still detect it?

Think it through — no sim needed (the sim's scenes are optical and radar):

The Sun's heating — plus the satellite's own powered electronics — keeps the object at tens of degrees, and when it slips into Earth's shadow it cools only slowly, so it keeps glowing in the infrared the whole time. Ask yourself: does *detecting* that glow require the Sun to be shining on the object right now? That is the point of this exercise.

What to look for: thermal-IR detects the object's own **emitted** heat, which — unlike reflected sunlight — does not require the Sun to be on the object at that moment. So it keeps working precisely where optical fails: **on objects inside Earth's shadow**. It is the passive complement to reflected-light optical, at the cost of cooled detectors and a bright IR sky.

While you watch, think about:

- What is the physical difference between reflected-sunlight optical and thermal-IR emission?
- Why does a thermal-IR sensor often need to be cooled?
- Between radar, reflected-light optical, and thermal-IR, which combination would you choose to never lose custody, and why?

Check your understanding: How can a passive sensor detect a satellite that is in Earth's shadow (no sunlight)?

- It cannot — being in shadow blocks every form of detection there is
- Thermal infrared: the object radiates its own heat regardless of sunlight
- By using a larger telescope with greater light-gathering power
- By waiting until the Moon is bright enough to illuminate the object

PART B · RA/DEC: pinning a direction on the sky

Optical gives angles — two precise coordinates and nothing else. (Sim scene: RA / DEC coordinates.)

Exercise 4 — Optical telescopes measure angles — RA and DEC

A telescope image pins an object's **direction** on the sky with great precision, but tells you **nothing about its distance** from one image alone (a nearby satellite and a far star can sit in the same direction). Astronomers name that direction with two angles that are literally **Earth's latitude and longitude projected straight up onto the sky** — extend the equator outward and it becomes the **celestial equator**; extend the poles and they become the **celestial poles**. On that projected grid, **Declination (Dec)** is the north-south angle (the sky's **latitude**) and **Right Ascension (RA)** is the east-west angle (the sky's **longitude**).

So an optical observation is a very accurate **pointing**: "the object was in *this* direction (RA, DEC) at *this* instant." It is the exact complement of radar. **Optical gives the object's precise direction on the sky — its angles, RA and DEC — but not its range** (its distance from the observer). **Radar gives a precise range but only a crude direction**. Angle without distance; distance without angle.

Predict first: From a single telescope image of a point of light, can you tell how FAR away it is? What CAN you tell precisely?

Try it in the simulator:

Open the **RA / DEC coordinates** scene. Drag to rotate the view, and move the **RA** and **Dec** sliders to see how the two angles pick out exactly one direction on the celestial sphere. Then consider a single image showing a dot at that measured spot: what do you know precisely, and what do you not know at all?

What to look for: one image gives a precise DIRECTION (RA and DEC) but no distance — a close satellite and a distant star can share the same line of sight. Range needs a second measurement (parallax, or radar).

While you watch, think about:

- RA/Dec are Earth's latitude/longitude projected up onto the sky — so what on the sky corresponds to the Earth's equator? To the poles?
- Why can't a single image distinguish a nearby satellite from a faraway star in the same direction?
- What extra information would you need to turn that direction into a 3-D position?

Check your understanding: What does a single optical image measure precisely — and what does it miss?

- It measures distance precisely, but says nothing about direction
- It measures direction — RA and DEC — precisely, but gives no distance
- It measures the object's speed directly from a single exposure
- It measures both distance and direction from the one image

Exercise 5 — Arcminutes and arcseconds

Positions on the sky are **angles**, and we need units far finer than a degree. One full circle is 360° , one degree is 60 **arcminutes** (′), and one arcminute is 60 **arcseconds** (″) — so one degree is 3,600 arcseconds.

These are tiny: the full Moon is about $\frac{1}{2}^\circ = 30' = 1,800''$ across, and a tracked satellite's position is often pinned to a few arcseconds. Precise angles are the whole currency of optical astrometry.

The simulator reports RA and Dec in **decimal degrees** (e.g. RA 60.0° , Dec $+30.0^\circ$) — clean for computation. But you will also see the traditional **sexagesimal** form: Dec as degrees:arcminutes:arcseconds (e.g. $+30^\circ 15' 22''$), and RA sometimes as **hours:minutes:seconds (HH MM SS.SS)**, dividing the 360° circle into 24 hours (so 1 hour of RA = 15°). Same angle, two notations — the degrees↔arcmin↔arcsec ladder ($\times 60$ each step) is what converts between them.

Predict first: The full Moon is about $\frac{1}{2}$ degree across. In arcseconds, is that closer to 30, 1,800, or 100,000?

Try it in the simulator:

Read carefully — this is key vocabulary. We measure positions in the sky as ANGLES.

What to look for: 1 full circle = 360 degrees. 1 degree = 60 arcminutes (′). 1 arcminute = 60 arcseconds (″). So 1 degree = 3,600 arcseconds. (The full Moon $\approx \frac{1}{2}^\circ = 30' = 1,800''$ across.)

While you watch, think about:

- Why do we need units as fine as arcseconds to track satellites?
- How many arcseconds is the full Moon's $\frac{1}{2}^\circ$ -diameter?
- A satellite pinned to "a few arcseconds" — how small a slice of the sky is that?

Check your understanding: How many arcseconds are in one degree?

- 60
- 360
- 3,600
- 86,400

Exercise 6 — Don't confuse arcseconds with seconds of TIME

A dangerous trap: "arcsecond" and "second" sound alike but measure completely different things. A **second of time** is 1/86,400 of a day. An **arcsecond** is 1/1,296,000 of a full circle — a measure of **angle**, not time.

They are linked by one memorable fact: at the **celestial equator** (Dec 0°) a star drifts past a fixed telescope at about **15 arcseconds of angle per second of time** (the sky turns $360^\circ = 1,296,000''$ in $\sim 86,400$ s). The number ties the two units together, but they remain different quantities.

Crucially the drift rate **depends on declination**. RA and Dec are **fixed to the sky**; the Earth spins about the polar axis, so the sight-line of a telescope **bolted in place** — parked, neither tracking nor being re-pointed — **sweeps through RA but not Dec**. (An operating telescope of course slews in both axes to *find* a target and then tracks; the point here is what Earth's spin alone does to a fixed sight-line.) For a star near the celestial **equator** that sweep is the full 15"/s; toward the **pole** the RA circles shrink, so the on-sky drift slows — and **at the celestial pole a star barely moves at all** (it just pivots in place). Formally the on-sky drift $\approx 15''/\text{s} \times \cos(\text{Dec})$. Note the two rates in play: every star crosses *RA coordinate lines* at the same 15°-per-hour clock rate, but the *physical arc it sweeps on the sky* — what streaks your image — shrinks with $\cos(\text{Dec})$.

Side note — so where do we put telescopes? A telescope on Earth's equator would enjoy the whole celestial sphere over a year and the solar-system planets passing near overhead — and a few historic observatories sit there (Quito, Ecuador, hosts one). But modern site committees avoid the equatorial belt: it is where tropical moisture, cloud and turbulence live, which wrecks both clear-sky hours and image sharpness ("seeing"). Instead they choose high, dry mountains at mid-latitudes — the Rubin Observatory on Cerro Pachón, Chile (latitude $\approx -30^\circ$, 2,650 m) and Keck on Mauna Kea, Hawaii ($\approx +20^\circ$, 4,145 m). And no modern telescope simply lets the sky streak by: whatever its latitude, it *tracks*, driving its mount to cancel the $15''/\text{s} \times \cos(\text{Dec})$ drift so stars sit still during the exposure. Space telescopes like Hubble skip the atmosphere altogether.

Predict first: A fixed telescope watches a star drift by. In one second of TIME, roughly how many arcseconds of ANGLE does it move — 1, 15, or 3,600?

Think it through — no sim needed:

Think about the two very different meanings of "second."

What to look for: a SECOND OF TIME is 1/86,400 of a day; an ARCSECOND is 1/1,296,000 of a full circle — an angle, not a duration. A fixed telescope sees a star near the celestial equator drift at about 15 arcseconds of angle per second of time (slower at higher declination, by $\cos \text{Dec}$).

While you watch, think about:

- Why does the sky drift at ~ 15 arcseconds per second of time at the equator? (Hint: 360° in ~ 24 h.)

- Why does a star near the celestial POLE barely move, while one on the equator races by? (Hint: RA circles shrink toward the pole.)
- The Earth spins about the polar axis — why does a ground telescope's sight-line sweep through RA but keep the same Dec?

Check your understanding: "Arcseconds" and "seconds" sound alike but mean different things. Which statement is correct?

- They are the same unit, used interchangeably by different fields
- An arcsecond is an angle, 1/3600 of a degree; a second is time
- An arcsecond is equal to sixty seconds of time, by definition
- Both are measures of time, differing only in their magnitude

Exercise 7 — Parallax — recovering the missing range

Optical gives direction but not distance. **Parallax** fixes that: view the object from **two vantage points** (two telescopes far apart, or one telescope at two times as the Earth carries it along), and the object's apparent direction **shifts** against the far background. A **nearby** object shifts a lot; a **distant** one barely shifts. The size of the shift, plus the known separation of the two viewpoints, gives the distance by simple triangulation.

This is the same trick your two eyes use for depth perception, and the same one astronomers use to measure distances to nearby stars.

Predict first: Before you try it: hold up **one finger** at arm's length. Predict what the finger will do against the far wall as you alternate eyes — and what will change if you bring it closer to your nose.

Try it right where you sit — no equipment, no sim:

Alternate which eye is open (close one, then the other, back and forth) and watch the finger against the distant background. Then bring the finger close to your nose and do it again.

What to look for: the finger **shifts sideways** against the distant background as you switch eyes — and it shifts **more** when it's closer. Your two eyes are two vantage points a few cm apart; the shift is **parallax**, and its size encodes distance. The **same thing happens when we view a satellite against the distant stars from two separated places** (or one place at two times): the satellite shifts against the star background, and that shift gives its range (its distance from the observer).

While you watch, think about:

- Why does a closer object show a bigger parallax shift than a distant one?
- What two things must you know to turn a parallax shift into an actual distance?
- Why does parallax to a distant cislunar object need telescopes very far apart (or a long time baseline)?

Check your understanding: How does parallax let optical sensors recover an object's distance?

- By measuring how bright the object appears and inferring range
- By viewing from two separated points and measuring the apparent shift
- By timing how long a radar echo takes to return from the object

- By reading off its RA and DEC from a single exposure

PART C • Take a picture: streaks are data

What a tracking telescope actually records, and what the ends of a streak mean. (Sim scene: Take a picture.)

Exercise 8 — The sky wheels overhead — how fast, and when does it repeat?

Open the **Take a picture** scene in the simulator and switch the mount to “**mount tracks a SATELLITE**”. (In the default “**mount tracks the STARS**” mode the mount holds the stars fixed, so only the satellite drifts.) Now watch the field drift: with the convention **North up, East left**, the stars slide **left→right** across the frame as the Earth turns beneath them. This is the same daily motion that makes the Sun rise and set.

The Earth spins once on its axis relative to the stars every **~24 hours** (the sidereal day is 23 h 56 m; the solar day is exactly 24 h). So a given patch of sky — the **same group of stars** — sweeps across your field and comes back around to the same place about **one day later**.

Side note — why is East on the LEFT? A map of the ground is drawn looking **down** at the Earth from outside: North up, East right. A sky chart is the opposite — you are looking **up** at the celestial sphere from *inside* it, and that flip of viewpoint mirror-reverses east and west. Hold a ground map over your head, printed side down, and its East swings to your left: that is exactly the view every telescope image records, which is why astronomers’ standard image convention is **North up, East left**.

Predict first: The stars drift steadily across the exposure. If you watched the very same star pattern leave the frame, roughly how long until that same pattern returns to the same spot?

Try it in the simulator:

1. In the simulator’s **Take a picture** scene, click “**mount tracks a SATELLITE**” — the default, “**mount tracks the STARS**”, holds the star field still (the satellite drifts instead).
2. Now watch the star field drift left→right. Note the direction (N up, E left) and the steady rate.

What to look for: the sky makes one full turn in about **24 hours**, so the same stars reappear in the same place roughly a day later. A geostationary satellite, by contrast, holds its spot against the ground — so relative to the drifting stars IT appears to move, which is exactly why it streaks (or the stars do, depending on what you track).

While you watch, think about:

- Why do the stars drift east-to-west across the sky over a night?
- How does a GEO satellite’s apparent motion against the stars differ from a star’s motion?
- If you image the same field 24 hours apart, what stays fixed and what has moved?

Check your understanding: Watching the star field drift, about how long until the same group of stars returns to the same place in the sky?

- About 1 hour, roughly the time a satellite takes to cross the sky
- About 90 minutes, matching the period of a low Earth orbit
- About 24 hours, which is one full rotation of the Earth
- About 27 days, matching the orbital period of the Moon

Exercise 9 — How do you measure a streaked object's position?

A tracking sensor takes a **time exposure**. If the telescope holds still, stars are points and a moving satellite draws a **streak**; if the telescope tracks the satellite, the satellite is a point and the **stars** streak. Either way, a streak is not a failure — it is data.

Analysts fit the streak's **two endpoints**: one is the object's position at shutter-**open**, the other at shutter-**close**, each tagged with a precise time. From one image that yields **two** timed positions plus the direction of travel — a compact packet of orbit information.

Predict first: A time-exposure shows a satellite as a streak, not a dot. Is that streak useless, or can you get a precise position (and motion) from it?

Try it in the simulator:

Look at a time-exposure image with a streaked object; think about what the two ends of the streak represent.

What to look for: a streak records where the object was over the whole exposure. The two endpoints are the positions at shutter-open and shutter-close, each with a precise time — two timed measurements from one image, plus the direction of motion.

While you watch, think about:

- Why do you get MORE information from a streak than from a single dot?
- What has to be recorded precisely for the endpoints to be useful?
- When would you deliberately let the satellite streak vs. track it into a point?

Check your understanding: You have a time-exposure with a streaked object. What's the best way to get a precise position (and even its motion) from it?

- You cannot — a streaked image is useless for measuring position
- Measure both endpoints, each tagged with the shutter open and close time
- Measure only the single brightest pixel along the length of the streak
- Average the whole streak together into one representative blob

Exercise 10 — Can you see its solar panels from the ground?

Resolving an object's **shape** is far harder than detecting its position. Two things fight you: the object's tiny angular size, and the atmosphere. A 30 m satellite at GEO (36,000 km) spans only about **0.2 arcseconds** on the sky — and the turbulent atmosphere blurs any ground image to about **1 arcsecond** of "seeing," no matter how big your telescope.

So from the ground a GEO satellite is an **unresolvable point of light**: you get its position and brightness, but not its shape. And giant mountaintop observatories don't escape this: Keck (Hawaii, 4,145 m) and the Rubin Observatory (Chile, ~2,650 m) sit above much of the weather but still *under* the turbulent air that sets the ~1" seeing, so they too record a GEO satellite as a point. Seeing components takes an inspector satellite flying close — **proximity operations** ("prox ops") — or a telescope above the atmosphere entirely.

Even that is no magic bullet: a Hubble-sized space telescope's diffraction limit ($\sim 0.05''$) spans about 9 m at GEO range, so a 30 m satellite would be at best a few blurry pixels. Getting *close* is what really resolves shape.

Predict first: A GEO satellite is ~ 30 m across at 36,000 km. Do you think a big enough ground telescope could make out its solar panels and dish?

Try it in the simulator:

Picture a GEO satellite (~ 30 m) at 36,000 km through a ground telescope; compare its $\sim 0.2''$ angular size to the $\sim 1''$ atmospheric blur.

What to look for: a 30 m satellite at 36,000 km spans only ~ 0.2 arcseconds. The atmosphere blurs everything to ~ 1 arcsecond, so from the ground it is an unresolvable point of light — you detect and track it, but you cannot see its shape.

While you watch, think about:

- Why can't a bigger telescope beat the atmospheric "seeing" limit from the ground?
- What would it take to actually image a GEO satellite's components?
- Why is "detecting/tracking" so much easier than "characterizing the shape"?

Check your understanding: Through a ground telescope, can you make out a GEO satellite's solar panels and dish?

- Yes, given a large enough telescope and a steady night
- No — at $\sim 0.2''$ it sits far below the $\sim 1''$ atmospheric blur, so it is a point
- Yes, but only at night when the sky background is dark enough
- Only if the satellite is geostationary and therefore holding still

PART D • Fit an orbit from a handful of angles

From tagged streak endpoints to a full orbit. (Sim scene: Tag & fit.)

Exercise 11 — Tag & fit — how few angles pin down an orbit?

Open the simulator's **Tag & fit an orbit** scene. The telescope tracks an unknown satellite, so in each exposure the **stars streak** and the satellite is the one crisp **point**. Click that point and the plate-solver locks onto the star-streak centers and reads off the satellite's **time, RA and Dec** with a measurement uncertainty of $\pm \sigma \approx 0.05''$ (read "plus or minus sigma": σ is the standard deviation of the measurement error, so each position is good to about a twentieth of an arcsecond either way) — one angles-only **line of sight** from Earth, with no range at all.

Watch what each sight-line buys you in the right-hand panel. **One** gives only a direction. **Two** pin the orbit's **plane** (both rays and Earth's center share one plane) — but the plane only: the orbit's size and shape are still wide open. From **three** tags on, the fit is over-determined, so the tiny measurement scatter becomes a finite **uncertainty band**. Now the key point — and the reason the status will NOT read DETERMINED after your third click: the band's **width is set by how far apart in time your tags are**. Tags bunched within minutes ride a short, barely-curved arc that many different orbits can match, so the band stays wide; tags spread across hours sweep out enough of the curve to collapse the band onto one orbit. Expect DETERMINED to take several tags, including some from *after* the time gap — the spread of your tags matters as much as their count.

Predict first: You have precise directions but zero range measurements. How many tagged exposures do you think it takes before the fit locks onto one orbit — 2, 3, or dozens?

Try it in the simulator:

1. In the **Tag & fit an orbit** scene, click the crisp point (labeled “← **the satellite (click me)**”) to plate-solve exposure 1, then press **next ▶** and tag exposure 2. Read the caption: two sight-lines fix the orbit’s **plane**.
2. Tag exposure 3 — still within the first tight cluster of times. The status flips to an orbit, but the translucent gold **uncertainty band** stays wide.
3. Keep pressing **next ▶** across the gap to the later exposures ($t + 2.7$ h onward) and tag them. Watch the band tighten and the chip flip to **STATUS: DETERMINED** □, with TLE-style elements and $\pm 1\sigma$ error bars. (⏮ **start over** resets if you want to retry.)

What to look for: a handful of angles-only observations pins down the entire orbit — no range measurement ever taken. Two tags fix the plane; three-plus over-determine it. But three tags bunched within ~16 minutes leave a wide uncertainty band — over a short arc many different orbits fit the same few directions — while tags spread across a few hours collapse the band onto one orbit with tight error bars.

While you watch, think about:

- Why do two sight-lines fix the orbit’s PLANE but not its size?
- Why does a short arc leave so much ambiguity, and why does spreading the tags across the night remove it?
- The recovered elements come with $\pm 1\sigma$ error bars. Where does that uncertainty come from, physically?

Check your understanding: In the Tag & fit scene, you tag three exposures taken within ~16 minutes of each other. What does the fit look like?

- It fails outright — angles-only data can never determine an orbit
- It locks immediately to tight error bars, since three tags are sufficient
- It cannot even establish which plane the object is orbiting in
- An orbit is found but with wide uncertainty — a short arc fits many orbits

PART E · Radar: later, fainter, dug out by averaging

One exercise, three facts: echoes come back later, drastically fainter, and averaging digs them out. (Sim scene: Radar sim.)

Exercise 12 — Radar measures range and range-rate

Because radar times its echo’s round trip, it measures the target’s **range** (distance) directly and very precisely: $\text{distance} = (\text{speed of light} \times \text{round-trip time}) \div 2$. And from the **Doppler shift** of the echo — the same effect that raises a siren’s pitch as it approaches — it measures the **range-rate**, how fast the distance is changing.

What radar does **not** do well is angle. A radar dish gives only a coarse pointing direction, so its cross-range (sideways) position is fuzzy. Radar’s picture of a target is “very sure how far and how fast toward/away, unsure where sideways.”

Predict first: A radar pings a satellite. Which does it nail down precisely — how FAR the object is, or its exact DIRECTION on the sky?

Think it through — no sim needed:

Recall that radar times the echo and reads its Doppler shift. Ask what each of those gives.

What to look for: radar gives precise range (from echo travel time) and range-rate (from Doppler), but only crude angular direction. It is the mirror image of optical.

While you watch, think about:

- How does timing the echo give distance? What role does the speed of light play?
- How does the Doppler shift of the echo reveal whether the object is approaching or receding, and how fast?
- Why is a radar's sideways (cross-range) position so much less precise than its range?

Check your understanding: What does a radar measure best?

- The object's exact direction on the sky, to fractions of an arcsecond
- Range and range-rate precisely, with only crude angular information
- The object's colour and physical shape from the returned waveform
- Its six orbital elements directly, without any further processing

Exercise 13 — Radar in one exercise: later, fainter, and dug out by averaging

A radar shouts a radio pulse and times the faint echo. The timing **is** the measurement: $\text{range} = c \cdot t / 2$, where **c** is the speed of light and **t** is the pulse's round-trip travel time (the $\div 2$ because **t** covers the trip out *and* back). And because the pulse spreads on the way out ($1/\text{range}^2$) and the echo spreads again on the way back ($1/\text{range}^2$), received power falls as **$1/\text{range}^4$** — double the distance, $16\times$ fainter; GEO at $\sim 53\times$ LEO's range returns $53^4 \approx 7.7 \text{ million}\times$ weaker. That one law is why radar owns LEO, strains at GEO, and cedes cislunar space to optical.

Radar's compensating strength: it is **active** — it transmits its own radio-frequency pulses and detects the reflected echo, so it needs no sunlight and works on targets deep in Earth's shadow, where a telescope (which needs reflected *sunlight*) goes blind.

Predict first: You fire one pulse at three targets — LEO (800 km), MEO (20,000 km), GEO (42,164 km). Two predictions: which echo arrives back first, and roughly how much weaker is GEO's than LEO's — $53\times$, 53^2 , or 53^4 ?

Try it in the simulator:

1. Open the Radar sim scene and press **FIRE PULSE** once. Watch the sweep dot cross the scope carrying its clock — the x-axis is **time since the pulse was sent**. LEO's bump appears almost at once (~ 5 ms); GEO's bin isn't reached until ~ 281 ms. Check the radar's arithmetic: $\text{range} = c \cdot t / 2 = (300,000 \text{ km/s} \times 0.281 \text{ s}) \div 2 \approx 42,000 \text{ km}$.
2. Now look at what landed: LEO towers over the trace, MEO is modest, and at GEO's bin there is... **grass**. GEO's echo is real but sits below the noise — a bump smaller than the wiggles it hides in. (Press **linear power view** to feel the $7.7\text{-million}\times$ ratio.)
3. Slide **integrate N pulses** — **N** is the number of pulses whose echoes get averaged together — to several hundred and press ► **START AVERAGING**. The radar fires

a stream of pulses; the echo repeats in the same time bin and adds up while the random noise partly cancels, so the grass calms and sinks — the **signal-to-noise ratio (SNR)**, how far the echo stands above the noise, grows as $\sqrt{N}$ — until GEO's bump stands clear.

What to look for: farther targets return **later** (time = $2 \cdot \text{range}/c$) and **drastically fainter** (range⁴). No amplifier can rescue a buried echo — it lifts signal and noise by the same factor — but **averaging** can. The echo adds **coherently**: the target sits at an essentially fixed range, so every pulse's echo returns after the *same* travel time and lands in the *same* time bin — N pulses stack the signal **N-fold**. The noise adds **incoherently**: it is random from pulse to pulse, sometimes up, sometimes down, so it only accumulates as $\sqrt{N}$. The ratio therefore improves as $N/\sqrt{N} = \sqrt{N}$: 100 pulses buy $\sqrt{100} = 10\times$ the SNR.

While you watch, think about:

- Where does integration's $\sqrt{N}$ improvement come from — what adds up, and what cancels?
- A satellite passes into Earth's shadow. Which sensor loses it — radar or optical — and why?
- Integration costs time staring at one target. What does that trade against with thousands of objects to track?

Check your understanding: A radar's received echo power scales with range as...

- $1/\text{range}$, falling off linearly as the target moves away
- $1/\text{range}^2$, the way light spreads from a distant star
- $1/\text{range}^4$ — the trip out and the trip back each cost range²
- Not at all — the echo strength is independent of range

PART F · Custody, maneuvers & intent

Finding it once is the easy part — the synthesis of everything above.

Exercise 14 — Custody and cadence — why you must keep looking

Finding an object once is easy; **keeping** it — maintaining **custody** — is the hard part, because every prediction goes stale. How fast it goes stale sets the **cadence**: how often you must re-observe. A well-behaved LEO orbit stays predictable for a long while; the chaotic cislunar regime (Module 6) goes stale in days, demanding frequent re-looks.

Custody is fundamentally a resource problem: you have limited sensors and thousands of objects, so you re-observe each just often enough to not lose it, spending scarce "looks" where they matter most.

Predict first: You have one LEO satellite and one chaotic cislunar object. Which one do you have to re-observe more often to keep from losing it, and why?

Think it through — no sim needed (Module 6 is the reference):

Recall from Module 6 how fast small position errors grow in chaotic cislunar orbits versus stable LEO orbits.

What to look for: the cislunar object goes stale far faster (chaotic divergence), so it needs a much brisker re-observation cadence; a stable LEO orbit can be predicted well

ahead and re-checked less often. Custody is about matching cadence to how fast predictions decay.

While you watch, think about:

- What sets how often you **MUST** re-observe a given object?
- Why does chaotic cislunar space make custody so much harder than LEO?
- With limited sensors and thousands of objects, how would you decide where to spend your "looks"?

Check your understanding: What determines the cadence at which you must re-observe an object to keep custody?

- Its altitude — higher orbits move through more space, so they always need more frequent looks
- How fast the prediction goes stale — chaotic orbits need frequent looks
- The sensor's aperture — a big enough telescope can safely wait much longer between looks
- How recently it was catalogued — newly added objects need brisk cadence permanently

Exercise 15 — Maneuver detection — it's not where it should be

Comparing a fresh observation against the **predicted** position from the object's known orbit is also how you catch a **maneuver**. If the object shows up where its old orbit says it should be, nothing changed. If it is **off** — beyond your measurement error — it must have **burned**: something changed its velocity.

This is why custody and maneuver detection are the same activity. The faster and more accurately you can re-observe, the smaller a maneuver you can catch, and the sooner. A rare, uncooperative object that quietly maneuvers between looks is the hardest case.

Predict first: You re-observe a satellite and it is 50 km from where its known orbit predicted. Assuming your measurement is good, what most likely happened?

Try it in the simulator:

Compare a fresh position measurement against the orbit's prediction. Ask what a significant, real discrepancy implies.

What to look for: a real position discrepancy (beyond measurement error) means the object's velocity changed — it maneuvered (burned). Maneuver detection is just custody plus comparison against prediction; better cadence and accuracy catch smaller, earlier maneuvers.

While you watch, think about:

- Why does a position that disagrees with the prediction imply a velocity change occurred?
- How do better cadence and better accuracy let you detect smaller maneuvers sooner?
- Why is an uncooperative object that maneuvers between your looks the hardest to keep custody of?

Check your understanding: You re-observe an object well away from its predicted position (beyond your errors). What happened?

- Its orbital plane precessed under J2 — the slow drift from Module 3 finally moved it
- It maneuvered — a burn changed its velocity and it left the old orbit
- A glint from a different face of the spacecraft shifted the measured image centroid
- Your observatory clock was off by a couple of seconds, displacing the whole measurement

Exercise 16 — Cooperative vs. uncooperative — and the ladder of knowing

Objects come in two flavors for tracking. A **cooperative** object helps you — it carries a transponder or beacon, or reports its own GPS position, so you know exactly where it is. An **uncooperative** object does not: a piece of debris, a dead satellite, or an adversary's craft that stays silent. Everything hard about tracking is about the uncooperative ones.

There is also a ladder of increasingly ambitious questions about a spacecraft's physical nature, capabilities and objectives — each rung demanding more and better data than the one below.

STEP 1 — Orbitology answers *where is it and where is it going*: a track.

STEP 2 — Characterization answers *what is it* — size, shape, spin, is there a robotic arm.

STEP 3 — Inferring intent is the hardest: *what is it doing and why* — is that approach an inspection, a refuel, or a threat?

Predict first: Rank these three questions from easiest to hardest to answer about an unknown satellite: "where is it going?", "what is it?", "what is it trying to do?"

Try it in the simulator:

Consider a silent, unknown object. Think about how much (and what kind of) data each rung of the ladder demands.

What to look for: orbitology (where/where-going) is the most tractable — the easiest to determine; a track suffices. Characterization (what is it) needs resolution or signatures. Inferring intent (what is it doing/why) is hardest, requiring characterization plus context and judgment. Uncooperative objects make every rung harder.

While you watch, think about:

- What is the difference between a cooperative and an uncooperative object, and why does it matter for tracking?
- Why is "what is its orbit" so much easier than "what is it trying to do"?
- What kinds of evidence would you need to responsibly infer intent (and why is that dangerous to get wrong)?

Check your understanding: Order the ladder of space-object knowledge from most tractable (easiest to determine) to least tractable (hardest to determine):

- Intent first, then characterization, and orbitology last of the three
- Orbitology, then characterization, then intent — hardest last
- They are all about equally tractable given the same observations
- Characterization first, then orbitology, with intent in the middle

Exercise 17 — Surveys vs. custody, and the neighborhood watch

Two complementary observing strategies. A wide **all-sky survey** sweeps the whole visible sky repeatedly, cataloging everything that moves — great for discovery and for not losing track of the population, but it can't stare at any one thing for long. A **pointed custody observation** does the opposite: it locks onto the few objects that matter and watches them closely.

A modern example: a **camera stationed in GEO** can watch its neighbors — a "**neighborhood watch**" — tracking nearby satellites against the star background, catching close approaches and **proximity operations** (one craft maneuvering near another to inspect, dock, or interfere). Surveys find the population; custody and neighborhood-watch sensors characterize the few that warrant a hard look.

Predict first: You must both (a) keep a catalog of everything in orbit and (b) watch one suspicious satellite closely. Can one sensor do both well at once? Why or why not?

Try it in the simulator:

Contrast a wide survey that scans the whole sky with a pointed sensor that stares at one target.

What to look for: no single mode does both well: surveys cover the whole population but can't dwell; pointed custody dwells on a few but can't cover everything. You need both — surveys to discover and catalog, custody/neighborhood-watch to characterize and infer intent.

While you watch, think about:

- What does an all-sky survey do well, and what does it sacrifice?
- Why is a camera stationed IN GEO well placed to watch its neighbors?
- How do surveys and pointed custody feed each other (discovery → hand-off → close watch)?

Check your understanding: How do wide surveys and pointed custody observations relate?

- They are two names for the same observing activity
- Surveys catalogue the whole population; pointed custody stares at a few
- Surveys are always the better use of a sensor's available time
- Custody observations remove any need to run wide surveys at all

Key points

- There are three ways to sense a space object: bounce **radar** off it, collect **reflected sunlight** (optical), or collect its emitted **thermal infrared**.
- **Active radar** transmits its own radio pulses and times the echo (works day/night, in shadow) but the echo falls as **range⁴** — so it owns LEO, strains at GEO, and mostly fails at cislunar range.
- **Optical** is passive: it only sees **sunlit** objects against a dark sky (the truck-headlights analogy), so it goes blind in Earth's **shadow** — where **thermal-IR** (self-emitted heat) can still detect the object.

- Radar measures **range and range-rate** (with crude angles); optical measures **precise angles (RA/DEC)** but no range. **Parallax** — the apparent shift from two viewpoints (the finger-and-eyes demo) — recovers the missing range.
- From the ground, distant satellites are **unresolvable points** (atmospheric seeing $\sim 1''$), so you get position and brightness, not shape; a streak's **two timed endpoints** pack extra orbit information into one image.
- Angles are measured in **arcminutes and arcseconds** ($1^\circ = 3,600''$) — not to be confused with **seconds of time**.
- Keeping **custody** means re-observing at a **cadence** set by how fast predictions decay (chaotic cislunar orbits need frequent looks); comparing to prediction is how you **detect maneuvers**.
- The ladder of knowing runs **orbitology** → **characterization** → **intent**, and it is far harder for **uncooperative** objects. Wide **surveys** catalog the population; pointed **custody** and a GEO "**neighborhood watch**" (proximity ops) characterize the few that matter.

Final quiz

Answers are checked in the online version.

Q1. A radar's received echo power scales with target range as...

- $1/\text{range}$
- $1/\text{range}^2$
- $1/\text{range}^4$
- independent of range

Q2. Why is optical tracking limited to sunlit objects against a dark sky?

- Optical telescopes can only be operated in the middle of the day
- It is passive — it sees only reflected sunlight, so the target must be lit
- Satellites emit their own visible light, which daylight then swamps
- Because of the range^4 law, which weakens the signal with distance

Q3. A satellite passes into Earth's shadow. Which passive sensor can still detect it?

- Reflected-light optical, which still picks up scattered earthshine
- Thermal infrared — the object radiates its own heat regardless of sunlight
- Neither one; a shadowed object cannot be detected passively at all
- Only radar, though radar is an active rather than a passive sensor

Q4 (select all that apply). Which of these affect the apparent magnitude (brightness) of a sunlit satellite?

- Illumination phase (Sun-satellite-observer angle)
- Orientation of the spacecraft bus
- Orientation of the solar panels (can throw a specular glint)
- Distance to the satellite
- Reflectivity (albedo) of the spacecraft components

Q5. What does a radar measure best, and what does an optical image measure best?

- Both measure distance well, differing only in their achievable precision
- Radar: range and range-rate, crude angles. Optical: precise angles, no range

- Both measure angles well, and neither recovers range from one look
- Radar measures colour and material; optical measures distance directly

Q6. A single telescope image gives a precise direction but no distance. How is the distance recovered?

- By how bright the object appears against the background sky
- By parallax — the apparent shift seen from two separated viewpoints
- It can never be recovered from optical observations at all
- By reading the object's right ascension from a single exposure

Q7. How many arcseconds are in one degree?

- 60
- 360
- 3,600
- 86,400

Q8. You re-observe a tracked object well off its predicted position (beyond measurement error). What happened?

- Its plane precessed under J2, an effect no propagator can anticipate
- It maneuvered: a burn changed its velocity and it left the old orbit
- A glint from a different face shifted the measured image centroid
- Your observatory clock drifted, displacing the whole measurement

Q9. Why is keeping custody of a chaotic cislunar object harder than a LEO satellite?

- Cislunar objects are physically larger and so harder to characterise
- Its predictions go stale fast, so it needs a much brisker re-observation cadence
- There is effectively no gravity out there to constrain the motion
- Objects in low Earth orbit are invisible to the sensors used for custody

Q10. Order the "ladder of knowing" about a space object from most tractable (easiest to determine) to least tractable (hardest to determine):

- intent → characterization → orbitology
- orbitology → characterization → intent
- all equal
- characterization → orbitology → intent

Q11. How do wide all-sky surveys and pointed custody observations relate?

- They are identical activities, differing only in what they are called
- Surveys catalogue the whole population; pointed custody dwells on a few
- Surveys make dedicated custody observations entirely unnecessary
- Custody observations remove any need to run wide all-sky surveys

Learn more

- Radar equation (the range⁴ law) — Wikipedia — why the two-way echo falls as 1/range⁴
https://en.wikipedia.org/wiki/Radar_equation
- Space domain / situational awareness — Wikipedia
https://en.wikipedia.org/wiki/Space_situational_awareness

- Equatorial coordinates: Right Ascension & Declination — Wikipedia
https://en.wikipedia.org/wiki/Equatorial_coordinate_system
- Parallax (distance from apparent shift) — Wikipedia — the finger-and-eyes trick, applied to the sky
<https://en.wikipedia.org/wiki/Parallax>
- Astronomical seeing (the $\sim 1''$ atmospheric blur) — Wikipedia
https://en.wikipedia.org/wiki/Astronomical_seeing
- Thermal (black-body) radiation — Wikipedia
https://en.wikipedia.org/wiki/Thermal_radiation
- Doppler effect (radar range-rate) — Wikipedia
https://en.wikipedia.org/wiki/Doppler_effect
- Space rendezvous & proximity operations — Wikipedia
https://en.wikipedia.org/wiki/Space_rendezvous